PROJECT ITI — FINAL REPORT

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Predicting Candidate Job-Change Likelihood

1. Problem Statement

Companies investing in candidate training programs need to identify potential 'flight risks' early to target retention efforts and plan hiring efficiently. This project models job-change likelihood as a binary classification problem (target = 1 for job-seekers, target = 0 for stayers) using demographic, education, and employment data.

The dataset is imbalanced (~3:1 negative to positive ratio), driving key preprocessing, weighting, and threshold tuning decisions.

2. Dataset Description

The dataset contains 19,158 training rows across 13 feature columns. Key features include city development index, company size/type, experience, and training hours.

Column	Description	Type	% Missing
enrollee_id	Unique candidate identifier	int	0%
city	City code (123 unique cities)	categorical	0%
city_development_index	City development index (0–1)	numeric	0%
gender	Gender	categorical	23.5%
relevent_experience	Relevant experience flag	categorical	0%
enrolled_university	University enrollment status	categorical	2.0%
education_level	Education level (Primary → PhD)	ordinal	2.4%
major_discipline	Major discipline	categorical	14.7%
experience	Years of experience (<1 to >20)	ordinal	0.34%
company_size	Current employer size bucket	ordinal	31.0%
company_type	Current employer type	categorical	32.0%
last_new_job	Years since last job change	ordinal	2.2%
training_hours	Hours of training completed	numeric	0%
target	0 = staying, 1 = job-seeking	binary label	0%

3. Methodology

3.1 Preprocessing & Missing Value Strategy

Mode and KNN imputation failed because missingness in company_size (31%) and company_type (32%) was non-random. Categorical NaNs were converted into an explicit 'Unknown' category, preserving the predictive signal. Outliers in training_hours were capped at 185.5 (IQR upper bound). Categoricals were encoded via ordinal mapping ('Unknown' = -1), frequency encoding (city), or one-hot encoding.

3.2 Model Evaluation & Threshold Tuning

Models were evaluated using 5-fold Stratified CV. For the tuned XGBoost model, out-of-fold probability predictions were extracted to sweep decision cut-offs between 0.10 and 0.90, optimizing for Class 1 F1-score.

4. Results

4.1 Verified CV Performance & Threshold Tuning

Model / Configuration	CV ROC-AUC	Class 1 Precision	Class 1 Recall	Class 1 F1	Accuracy
Logistic Regression (Balanced)	0.7615	0.48	0.69	0.56	0.73
Random Forest (SMOTE)	0.8003	0.57	0.72	0.63	0.79
XGBoost (Default @ 0.50)	0.8008	0.5587	0.7454	0.6350	0.7850
XGBoost (SMOTE)	0.7901	0.5856	0.6808	0.6350	0.8078
Ensemble (3 Merged Models)	0.7918	0.5786	0.7189	0.6456	0.7974

4.2 XGBoost Optimized Confusion Matrix Analysis

At the optimal threshold of 0.52, the tuned XGBoost model correctly detected 3,461 flight-risk candidates (True Positives) while missing 1,316 (False Negatives). It accurately classified 11,719 non-job seekers (True Negatives) with 2,662 false alarms (False Positives), achieving 79% overall accuracy.

5. Conclusions

- Treating missing categorical values as 'Unknown' preserved essential candidate response signals.
- XGBoost paired with SMOTE achieved peak accuracy with 80%.
- Ensemble method was considered as the best due to the overall scores across the other measurements